EXAMPLE 4.24

Find the instantaneous value of alternating voltage $v = 10\sin(3\pi \times 10^4\, t)$ volt at i) 0 s ii) 50 μs iii) 75 μs.

Solution

The given equation is $v = 10\sin\left(3\pi \times 10^4 t\right)$

(i) At t = 0 s,

$$\upsilon = 10\sin 0^\circ = 0 \text{ V}$$

(ii) At t = 50 μs,

$$\begin{aligned} v &= 10\sin\left(3\pi \times 10^4 \times 50 \times 10^{-6}\right) \\ &= 10\sin\left(150\pi \times 10^{-2} \times \frac{180^\circ}{\pi}\right) \\ &= 10\sin(270^\circ) = 10 \times -1 \\ &= -10 \text{ V} \end{aligned}$$

(iii) At t = 75 μs,

$$\begin{aligned} v &= 10\sin\left(3\pi \times 10^4 \times 75 \times 10^{-6}\right) \\ &= 10\sin\left(225\pi \times 10^{-2} \times \frac{180^\circ}{\pi}\right) \\ &= 10\sin(405^\circ) = 10\sin 45^\circ \\ &= 10 \times {}^{1}\!/\!_{\sqrt{2}} = 7.07 \text{ V} \end{aligned}$$

EXAMPLE 4.25

The current in an inductive circuit is given by 0.3 sin (200t – 40°) A. Write the equation for the voltage across it if the inductance is 40 mH.

Solution

$L = 40 \times 10^{-3}$ H; $i = 0.3$ sin (200t – 40°)

$$X_L = \omega L = 200 \times 40 \times 10^{-3} = 8\ \Omega$$
$$V_m = I_m X_L = 0.3 \times 8 = 2.4 \text{ V}$$

In an inductive circuit, the voltage leads the current by 90°. Therefore,

$$v = V_m \sin(\omega t + 90^\circ)$$
$$v = 2.4\sin(200t - 40^\circ + 90^\circ)$$
$$v = 2.4\sin(200t + 50^\circ)\text{V}$$

4.8 POWER IN AC CIRCUITS

4.8.1 Introduction of power in AC circuits

Power of a circuit is defined as the rate of consumption of electric energy in that circuit. It is given by the product of the voltage and current. In an AC circuit, the voltage and current vary continuously with time. Let us first calculate the power at an instant and then it is averaged over a complete cycle.

The alternating voltage and alternating current in the series inductive *RLC* circuit at an instant are given by

$$v = V_m \sin\omega t \quad \text{and} \quad i = I_m \sin(\omega t + \phi)$$

where ϕ is the phase angle between v and i. The instantaneous power is then written as

$$\begin{aligned} P &= v\, i \\ &= V_m I_m \sin\omega t \sin(\omega t + \phi) \\ &= V_m I_m \sin\omega t \left[\sin\omega t \cos\phi + \cos\omega t \sin\phi\right] \\ P &= V_m I_m \left[\cos\phi \sin^2\omega t + \sin\omega t \cos\omega t \sin\phi\right] \end{aligned} \qquad (4.54)$$

Here the average of $\sin^2\omega t$ over a cycle is ${}^{1}\!/\!_{2}$ and that of $\sin\omega t \cos\omega t$ is zero. Substituting these values, we obtain average power over a cycle.

$$\begin{aligned} P_{av} &= V_m I_m \cos\phi \times {}^{1}\!/\!_{2} \\ &= \frac{V_m}{\sqrt{2}} \frac{I_m}{\sqrt{2}} \cos\phi \\ P_{av} &= V_{RMS}\, I_{RMS} \cos\phi \end{aligned} \qquad (4.55)$$

where $V_{RMS} I_{RMS}$ is called apparent power and cos ϕ is power factor. The average power of an AC circuit is also known as the true power of the circuit.

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